

Analysing the Data from YZU Portal Platform

Benjamin Kuo

*International Bachelor Program in Informatics
Yuan Ze University
Taoyuan, Taiwan*

Aman

*International Bachelor Program in Informatics
Yuan Ze University
Taoyuan, Taiwan*

Abstract—This project investigates whether people can find any useful information in the homework data from the YZU portal platform. We found out which weeks have highest loading by our code. In addition, we found that homework deadlines in one week would not result in more late submissions. Finally, we predict the future semester homework loading by the last semester, which use two models to implement.

Index Terms—data fetching, assignments, EDA, k-means, logistic regression, random forest, model evaluation

and combines those data to generate a “homeworks.json” for further investigation.

The json file includes attributes such as the semester of the course, course name, assignment name, deadline, late minutes for late submission, assignment type and score. However, the course name and the assignment type will not be used in this project since they are not important. They are only for helping people understand the data.

I. INTRODUCTION

In some students’ collage life, they often hang out with their friends, but at the same time, they have to handle their assignments. However, the deadline of assignments is not separated evenly. There are weeks that might be many tests and homework, which cause the workload to be crowded. The purpose of this project is to find the hidden patterns behind the deadline of homework, so students can avoid those busy weeks when they arrange the vacations before the deadline of homework is polished or announced.

By analyzing the homework data, we can get information including assignment name, course name, semester of the course, deadlines, late submissions and scores. Those data are useful to analyze which weeks are busy and the relationship between those attributes.

The goal of this project can be separated into three parts. First, we identify what high-workload weeks are. Second, we analyze whether more homework due days would lead to more late submissions. And third, we predict the future workload by the homework deadlines in previous semester.

II. DATA COLLECTION

The data of this project came from YZU portal platform, which is fetched by the python program we created. This platform contains the homework data for each student in Yuan Ze University. However, the password for one student can only access his or her homework records.

To collect the data, the code will get the password for the platform from a “.env” file, which allows it to login by itself. There is a section in the web page that student could find every course he or she attended. Those courses are sorted by the semesters, so the program need to clicking every button for each semesters to reveal course links which leads user to every course page. After visiting every link, it will collect every homework record chart, which include every assignment,

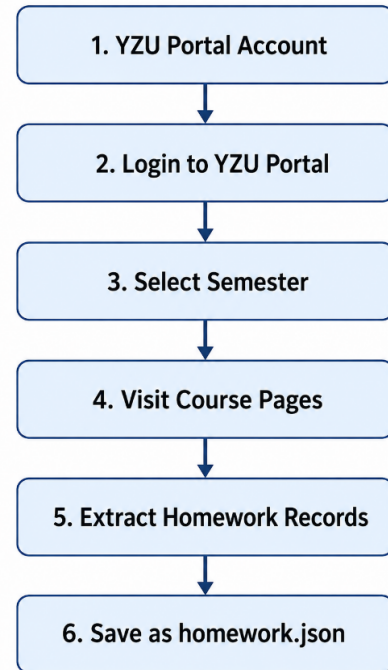


Fig. 1. Data collection workflow.

III. DATA PREPROCESSING

The purpose of preprocessing is because the data from “homeworks.json” might not be complete, and the data might in the wrong format. Therefore, we must clean it before any analysis. We implement it with a python notebook.

We clean the data with several steps. First, we change the data types into different formats so we can use them easily with the code. For instance, converting “late_minutes” into numeric data types. If data doesn’t have the information of deadline, we will delete them. Then, we handle the missing values. For example, adding 0 to data which don’t record

“late_minutes”. After that, we create the features for late-submission indicators and score-related indicators such as “is_late”, “has_score” and “is_low_score”. We also create time features which are “due_date”, “due_year”, “due_month”, “due_week”, “due_weekday”, “due_hour” and “year_week”. Last but not least, we cap the outliers of late_minutes by IQR, and create a new feature called “late_minutes_capped”.

TABLE I
DATA PREPROCESSING STEPS

Step	Process	Purpose
1	Convert data types	Convert dates and numbers
2	Remove invalid records	Drop records without deadline
3	Handle missing values	Fill late minutes with 0 and missing categories with Unknown
4	Create features	Late, score and time features
5	Handle outliers	Cap extremely late minutes using IQR

IV. EXPLORATORY DATA ANALYSIS

EDA is being used for analyze the distribution of homework. We create different indicators based on features, such as “homework_count”, “late_count”, “late_rate”, “avg_late_minutes” and “low_score_rate”, that tells us which weeks are high workload. Those indicators could be used in later sections. There are three ways to separate deadlines into three groups.

A. The Number of Deadlines in a Week

It is the most intuitive way to group those data. We found that in the record, the highest number of deadlines in a week is 8 (in the data set of Benjamin’s homework). The data is divided based on the 33rd and 66th percentiles of homework counts. This kind of division can avoid subjective classification. .

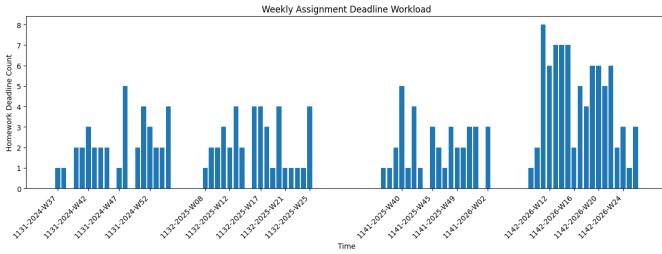


Fig. 2. Weekly homework deadline count across semesters.

B. Weighted Workload Score

We create a weighted function to evaluate the weekly workload. It includes three factors, “homework_count”, “late_rate” and “avg_late_minutes”.

$$\begin{aligned}
 \text{Workload} = & \text{homework count} \times 60\% \\
 & + \text{late rate} \times 25\% \\
 & + \text{avg late minutes} \times 15\%
 \end{aligned} \quad (1)$$

If the score is below 40 is low; if it is above 60 is high. Otherwise, it is medium.

C. The Cluster

By using the k-means, we can create three cluster that have similar attribute. We group them with the attributes like “homework_count”, “late_count”, “late_rate”, “avg_late_minutes” and “low_score_rate”. Then we assigned those group as “high assignment-count”, “medium assignment-count” or “low assignment-count” based on the average of homework in every week in the same cluster.

TABLE II
K-MEANS CLUSTER PROFILE

Group	Weeks	Avg HW	Late Rate	Late Min	Low Score
Low assign	13	1.46	0.85	3328.85	0.19
Med assign	37	2.73	0.00	0.00	0.13
High assign	17	4.47	0.38	546.69	0.10

Note: Avg HW = average homework count; Late Rate = average late-submission rate; Late Min = average late minutes; Low Score = average low-score rate.

D. Summary of EDA Findings

The result from EDA shows that homework isn’t spread evenly in every weeks. Counting deadlines in a week turn out to be the most direct and interpretable indicator since it is simple, stable and would not affect following modeling stage. The next research is about finding the relationship between workload and late submissions. Therefore, if an indicator is affected by any value related to late rate, it will not give us a reliable outcome.

V. DATA MINING METHODS

A. Grouping with Workload indicator

We group those deadlines into three groups, which is based on how many deadlines per week. To avoid subsection classification, we group those data based on 33rd and 66th percentiles of homework counts, which help us to do the analysis and prediction later.

B. Deadline Count and Late Rate Analysis

In this section, we compared the average homework count, average late rate, average late minutes and average score in different groups (workload: low, medium, high). However, we discovered that when there are more homework in a week, there are lower late rate. I believe it is because when there is only one homework in a week, if student forget to turn in, he or she would get a record of 100% late rate in that week.

C. High Workload Prediction

We tried to use Logistic Regression and Random Forest Classifier to predict the heavy workload timing with the assignment deadlines from last semester. We identify a week has over three homework (include 3) as a high workload week. Then, we put those data to those two model to and judge both model with four indicators, such as “accuracy”, “precision”, “recall” and “f1_score”. I adjust some parameters to avoid overfitting, which improve model “f1_score”.

VI. RESULTS AND DISCUSSION

The project result shows not only when is a busy week in a semester, but also indicates that there it is easier to get higher late rate when there is not much homework. In my opinion, when a student knows that there will be lots of work in future week, they might finished them earlier, which reduce the workload in that week. If there are not much homework in a week, student will be more likely to forget them.

For the future workload prediction, Logistic Regression got an F1-score 0.70, while Random Forest got F1-score 0.68. It shows that both models have acceptable performance. However, due to the size of the dataset, the result still limited.

VII. CONCLUSION

Our project shows that homework deadline data obtained from the YZU portal platform can be used to identify implicit patterns behind the deadlines. It supports to predict the future workload from previous semester, which have the best F1-score 0.7. This project shows that numbers of homework deadline in weeks are good indicator for defining whether a week is high-workload, whereas logistic regression and random forest are useful to predict future high-workload weeks. In the future, we can add more data, such as different students and more assignment attributes to improve the model prediction.